

Introduction to Industrial Management (Humanities III) (HSMC-501)

Course Details

Credit Points: 3

Syllabus Outline

Unit 1

Introduction. System- concept, definition, types, parameters, variables and behavior. Management - definition and functions. Organization structure: Definition. Goals. Factors considered in formulating structure. Types. Advantages and disadvantages. Applications. Concept, meaning and importance of division of labor, scalar & functional processes, span of control, delegation of authority, centralization and decentralization in industrial management.

Unit 2

Organizational culture and climate meaning, differences and factors affecting them. Moral-factors affecting moral. Relationship between moral and productivity. Job satisfaction- factors influencing job satisfaction. Important provisions of factory act and labor laws. Critical Path Method (CPM) and Programme Evaluation Review Technique (PERT): CPM & PERT-meaning, features, difference, applications. Understand different terms used in network diagram. Draw network diagram for a real life project containing 10-15 activities, computation of LPO and EPO. Determination of critical path on network. Floats, its types and determination of floats. Crashing of network, updating and its applications.

Unit 3

Materials Management: Material management-definition, functions, importance, relationship with other departments. Purchase - objectives, purchasing systems, purchase procedure, terms and forms used in purchase department. Storekeeping- functions, classification of stores as centralized and decentralized with their advantages, disadvantages and application in actual practice. Functions of store, types of records maintained by store, various types and applications of storage equipment, need and general methods for codification of stores. Inventory control: Definition. Objectives. Derivation for expression for Economic Order Quantity (EOQ) and numeric examples. ABC analysis and other modern methods of analysis. Various types of inventory models such as Wilson's inventory model, replenishment model and two bin model. Material Requirement Planning (MRP)- concept, applications and brief details about software packages available in market.

Unit 4

Production planning and Control (PPC): Types and examples of production. PPC: Need and importance. Functions. Forms used and their importance. General approach for each type of production. Scheduling-meaning and need for productivity and utilisation. Gantt chart- Format and method to prepare. Critical ratio scheduling-method and numeric examples. Scheduling using Gantt Chart. Bottlenecking- meaning, effect and ways to reduce.

Unit 5

Value Analysis (VA) and Cost Control: VA-definition, terms used, process and importance. VA flow diagram. DARSIRI method of VA. Case study of VA-at least two. Waste-types, sources and ways to reduce them. Cost control-methods and important guide lines.

Unit 6

Recent Trends in IM: ERP (Enterprise resource planning) - concept, features and applications. Important features of MS Project. Logistics- concept, need and benefits. Just in Time (JIT)-concept and benefits. Supply chain management-concept and benefits.